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THE TASK



CQ-DINO: Mitigating Gradient Dilution via Category Queries...
NeurIPS 2025 · arXiv 2503.18430



Match target
cheapest config that proves the paper’s claim

metric AP / AP50 / AP75
scope V3Det val · 29,821 imgs
target 46.3 / 51.5 / 48.4 ±5%

 configuration withheld



What the authors released

✓ data

✓ code

✓ weights

AVAILABILITY TIER

▶ Run execute what shipped

Retrain  weights withheld

Reimplement  code withheld too

the availability cliff
success falls as the release thins

 budget 8 H100-h

×100 papers

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THE RUN



Reproduction agent

deepseek-v4

workspace_bash

write_file

fetch_url

 run_gpu

run 2889575 · 107 rounds · 6.0 of 8 H100-h

PLAN · r0-4

● [plan] AP 46.3 on V3Det · design eval, run it

FETCH · r5-8

○ \$ git clone github.com/.../CQ-DINO

○ \$ gdown ...swinb1k_v3det.pth # 1.89 GB

BUILD · r9-61

○ \$ pip install mmcv==2.2.0

● ERROR: no aarch64 wheel

○ AH! no aarch64 wheel. build it from source

○ \$ FORCE_CUDA=1 pip install -e mmcv/

RUN · r62-64

● \$ run_gpu tools/test.py # 29,821 imgs GPU

● srun: task 0: Out Of Memory

RESCUE & SCORE · r65-92

○ OOM again, but the preds survived in /tmp

○ \$ cp /tmp/results.bbox.json evidence/

● \$ run_gpu chunked_eval.py # score them GPU

VERIFY & REPORT · r93-107

○ my scorer agrees with COCOeval, twice

○ \$ write_file report.json

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THE REPORT

report.json

claims: "reproduced"

evidence/ eval.log · commands.log
results.bbox.json · patches/
every number traces to a logged run

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THE AUDIT



Adversarial audit
re-derives the verdict from the evidence

✓ metric recomputed from raw predictions

✓ exact checkpoint, sibling rows ruled out

✓ AP_{S/M/L} values not in the paper: no echo

🚩 training arm untested (released ckpt only)

THE RUN CLAIMS
AP 46.3 ✓

vs

THE AUDIT FINDS
AP 46.3 ✓

RECLAIM scores the audited verdict

bbox_mAP 0.463 0.515 0.484 = paper’s 46.3 / 51.5 / 48.4 ✓

● plan ○ thought ○ step ● GPU run ● error \$ agent command